BEFORE ENGINE START

Familiarise with HANDBOOK first!
Preflight Inspection . . . . . COMPLETED
According to Handbook, Cpt. 4.3
Rudder Pedal Position . . . . . SET
Seat belts . . . . . TIGHTEN
Flight Controls . . . . . CHECKED
Canopy . . . . . LOCKED
Parking Break . . . . . SET
Apply pressure and set, then 'pump'.
For usage of external Power,
see notes in Handbook Cpt. 4.4.2

ENGINE START

Circuit Breakers . . . . . CHECKED
Backup Battery . . . . . ON
GTN650Xi, MFD, EFIS . . . . . BOOTING
MFD
Wait for boot sequence.
Check FUEL QTY!
Conf. amber "BACKUP BATT" msg.
Conf. CO value 255 ppm, then decr.
Master Switch . . . . . ON
MFD
Conf. amber "BACKUP BATT" msg.
CAS message OFF
Note down Total Hours from MFD!
Backup Battery . . . . . OFF
GEN A/B . . . . . ON
Voltage M . . . . . MIN. 12 V
(Batt/Alt-C Bus)
Fuel Selector . . . . . MORE EMPTY TANK
Propeller Control . . . . . FINE
Fully forward
Throttle . . . . . CLOSED
Strobe Light . . . . . ON
LANE A / LANE B . . . . . ON
Main Fuel Pump . . . . . ON
– Continues next column!

CONTINUE: ENGINE START

Start Power . . . . . ON (HOLD!)
LANE A/B Lamps:
wait till lamps turn OFF
MFD Engine Indication . . . . . CHECKED
→ active and correct
→ Fuel Pressure in Limits
Throttle Lever:
Coolant/Oil Tmp. <10 °C ≥ 55 %
Coolant/Oil Tmp. >10 °C ca. 45 %
% as indicated as "T" on MFD.
See HINT and Start Performance Chart
in Handbook Cpt. 4.4.3 on P. 4-8!
ENGINE START:
Hold "Start Power", THEN press "START"
until Engine fires. Then release
Start-Button and "Start Power".
Throttle . . . . . 2000 RPM
Oil Pressure . . . . . CHECKED
LANE A/B Indicator Lamps . . . . . OFF
Both LANE SWITCHES remain ON!
Only the Indicator Lamps should go out.
Throttle . . . . . 2500 RPM FOR 10 SEC
To engage/sync generators.
Then back to 2000 RPM
Voltage M / Voltage B . . . . . 13.4 / 14.0 V

AFTER ENGINE START

Avionics . . . . . ON
ALT-C . . . . . ON
Voltage M . . . . . CHECKED
Increasing, nom. 13.8 V
EFIS . . . . . ON
EFIS Power Source . . . . . CHECKED
Verify NOT on internal Battery
Baro PFD + MFD . . . . . SET
StbyEFIS, GTN are synced. MFD isn't!

BEFORE TAXIING

Nav Light (at night) . . . . . AS REQUIRED
Landing-/Taxi-Light . . . . . ON
A/C moving: Lights on!
PFD, stby EFIS, MFD . . . . . CHECKED
ADHRS stabilised

TAXIING

Keep RPM < 2500, taxiing with walking speed! Check BRAKES & STEERING! Ground Control CHECKED

Brakes . . . . . CHECKED
Indications . . . . . CHECKED
Attitude/direction: PFD, MFD, stby EFIS

ENGINE RUN UP

Parking Brake . . . . . SET
Landing-/Taxi-Light . . . . . AS REQUIRED
A/C standing still: Lights off!

Engine Warm-Up:
2500 RPM until oil temp > 50 °C.

Propeller Control . . . . . FINE
Fully forward

Pump and Fuel Supply Check
Throttle: 3000 RPM
AUX Pump: ON, pressuere in limits
AUX Pump Light: ON
Main Fuel Pump: OFF
Pressure still in limits: CHECKED
Main Fuel Pump: ON

Fuel Tank . . . . . SIDES SWITCHED
Fuel Pressure . . . . . CHECKED
Fuel Tank . . . . . FULLEST TANK
– Continues next column!

CONTINUE: ENGINE RUN UP

LANE Checks
Throttle: set to 2500 RPM
LANE A Switch: OFF
Max. vary ±250 RPM (aft. 15 sec.)
Check Engine Params in Limits!
LANE A Switch: ON (wait 3 sec.)
LANE B Switch: OFF
Max. vary ±250 RPM (aft. 15 sec.)
Check Engine Params in Limits!
LANE B Switch: ON (wait 3 sec.)

Propeller Check
Throttle: set to 4000 RPM
Propeller Control: CHECKED
Cycle 3 times. Move lever SLOWLY backwards, but fast forward.

Throttle . . . . . IDLE CHECK
Throttle . . . . . SET 3000 RPM

BEFORE TAKE-OFF

Instruments . . . . . CHECKED
Pitot-Heat . . . . . AS REQUIRED
Flight Controls . . . . . CHECKED
Cockpit Canopy . . . . . CLOSED & LOCKED
Seat Belts . . . . . TIGHTENED
AEPS System . . . . . UNLOCKED
Store pin in middle compartment!
Engine Instruments . . . . . CHECKED

Autopilot
Autopilot Switch: ON
Engage, for each test
Check if AP can be overpowered!
Check AP Disconnect by:
Any Trim input, AP-Disc. button.

Autopilot Switch . . . . . OFF
Autopilot Panel LEDs . . . . . ALL OFF
Trim . . . . . T/O POSITION
Flaps . . . . . AS REQUIRED
Flaps 0 recommended.
Flaps 1 also possible - See AFM!

TAKE-OFF

Landing-/Taxi-Light . . . . . ON
Brakes . . . . . RELEASED
Propeller Control . . . . . FULL FORWARD

When lined-up:
Throttle . . . . . FULL POWER

Engine Instruments:
RPM: min. 5700, max. 5800
Manifold Pressure: within LIMITS
Manifold Temp: Below 65 °C

Airspeed Indication . . . . . COMES ALIVE

TAKE-OFF SPEEDS
Nosewheel Unstick . . . . . 50 KIAS
Lift-Off . . . . . 60 KIAS
Initial Climb . . . . . 70 KIAS
Flaps retract . . . . . 75 KIAS
Not below 150 ft AGL!
Climb . . . . . V_Y 78 KIAS

CLIMB

Climbing Speeds
- v_y best ROC 78 KIAS
- v_x best AOC 63 KIAS

Throttle . . . . . FULL FORWARD

Prop Control
- Set Engine Speed
- max. 5800 RPM for 5 min
- max. continuous 5500 RPM

Instruments . . . . . T & P IN LIMITS
AUX Pump . . . . . OFF
Fuel Pressure . . . . . CHECKED
Landing Light . . . . . AS REQUIRED
Outside circuit or CTR: OFF

CRUISE

Fuel Selector
- Switch every 25 min, latest.
- Check for balanced quantity!
- See notes in AFM Cpt. 4.4.13.

Throttle . . . . . AS REQUIRED
Prop Control . . . . . MAX. 5500 RPM
Fuel Flow . . . . . CHECKED
Pitot Heat . . . . . AS REQUIRED

Changes in PWR:
Increase Power ⇑ . . . . . 1. RPM - 2. MAP
Decrease Power ⇓ . . . . . 1. MAP - 2. RPM

PWR	RPM	MAP
100 %	5500 RPM	40.5 inHg
75 %	5000 RPM	33.0 inHg
59 %	4500 RPM	25.5 inHg

DECENT

Throttle . . . . . 15...17 INHG
Prop Control . . . . . MAX. 5000 RPM
Airspeed . . . . . 90...100 KIAS
Pitot-Heat . . . . . AS REQUIRED
Do NOT reduce Throttle to IDLE
in cold OAT environments!

BEFORE LANDING

Seat Belt . . . . . TIGHTENED
Cockpit . . . . . PREPARED
Autopilot . . . . . DISENGAGED
Autopilot Switch . . . . . OFF
Throttle . . . . . AS REQUIRED
Speed . . . . . $\leq$ 83 KIAS
Flaps . . . . . FLAPS 1
Trim . . . . . AS REQUIRED
Landing Light . . . . . ON
AUX Pump . . . . . ON

Final Approach
Flaps . . . . . FLAPS 2
Speed . . . . . 65 KIAS
Trim . . . . . AS REQUIRED
Propeller Control . . . . . FINE
Fully forward
Throttle . . . . . SET
- Approx. 7 inHG,
- Maint. RPM > 2600

BALKED LANDING / GO-AROUND

Throttle . . . . . FULL FORWARD
Speed . . . . . 70 KIAS
Flaps . . . . . FLAPS 1
Trim . . . . . ADJUSTED
Climb . . . . . V_Y 78 KIAS
Follow T/O Procedures

LANDING

Throttle . . . . . CLOSED
Touch-down . . . . . MAIN WHEELS
Brakes . . . . . AS NEEDED

AFTER LANDING

Flaps . . . . . RETRACTED
Engine Speed . . . . . SET
Taxi with walking speed!
Landing-/Taxi-Light . . . . . ON
Pitot Heat . . . . . OFF
AUX Fuel Pump . . . . . OFF

SHUTDOWN

Parking Brake . . . . . SET
Landing-/Taxi-Light . . . . . OFF
Propeller Control . . . . . FINE
Engine Speed . . . . . IDLE
Instruments . . . . . ALL IN LIMITS
ALT-C . . . . . OFF
Check Volts M $\geq$ 13.4 V!
min. 2 min COOL-DOWN!
Use timer in PFD or MFD!
Note down Total Hours from MFD!
Lane A . . . . . OFF
after 2 sec, LAMP: . . . . . CHECKED ON
Lane B . . . . . OFF
Engine will STOP!
Main Fuel Pump . . . . . OFF
GEN A/B . . . . . OFF
Avionics . . . . . OFF
EFIS . . . . . OFF
Strobes . . . . . OFF
Master . . . . . OFF
AEPS Locking Pin . . . . . INSERTED
Parking Brake . . . . . AS REQUIRED
All switches are OFF . . . . . CHECKED

Parking and tie-down:
All Lane & Master Switches . . . . . OFF
Parking Brake . . . . . SET
For short time parking only,
for long time, use chocks!
Canopy . . . . . CLOSED & LOCKED
AIRCRAFT SECURED!